# AI CHATBOT AND RAG:

## AUTOMATED CONVERSATIONS AND DOCUMENT INTELLIGENCE

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# PROJECT 1: AI CHATBOT

AUTOMATED CONVERSATIONS

# OVERVIEW:

An automated, friendly and upbeat AI chat assistant named "Bubbles" built using Python and cloud-based AI technology.

# PROBLEM STATEMENT

**Rigid Systems:**

Traditional automated bots feel robotic, cannot understand natural speech, and forget what the user said just a moment ago.

**Disengaged Users:**

Dry, overly formal automated responses push younger audiences away.

# KEY FEATURES AND FUNCTIONALITIES

### Smart Conversation Memory

- The bot remembers what was said earlier in the chat.
- Customers don't have to repeat themselves when asking follow-up questions, making the chat feel like a natural conversation with a human.

### Engaging Brand Personality

- The bot uses a welcoming, upbeat tone with friendly, modern language and emojis.
- Breaks the ice and removes the cold, robotic feel of standard corporate bots, instantly boosting customer satisfaction.

### Lightning-Fast Responses

- Connected to high-speed cloud systems.
- Answers drop in less than a second, ensuring customers get immediate help without frustrating delays.

### Guardrails on Length & Style

- The system is programmed to keep answers brief and focused.
- Prevents the AI from rambling or overwhelming the user, delivering short, straight-to-the-point answers.

# MY CONTRIBUTION:

- **Built the Chatbot**: Wrote the underlying code that lets users type and talk to the AI system.
- **Designed the "Bubbles" Voice**: Programmed the exact guidelines to make the assistant sound warm, upbeat, and customer-friendly.
- **Gave the AI a Memory**: Set up the system so the bot remembers the conversation history and understands follow-up questions.
- **Fixed System Breaks**: Handled behind-the-scenes software upgrades to ensure the chatbot stays online and runs smoothly without crashing.

# CHALLENGES & FIXES

**Challenge:**

The original AI engine was suddenly discontinued by the outside provider, which broke the chatbot.

**Fix:**

I quickly stepped in and upgraded the system to use a brand-new, modern AI engine (llama-3.1-8b-instant), restoring full service immediately.

**Challenge:**

Early tests showed the AI could become too repetitive or wander off into overly long, rambling answers.

**Fix:**

I fine-tuned the system's control rules, strictly limiting message lengths and balancing creativity so answers stay short, fresh, and on-topic.

# PROJECT 2: RETRIEVAL-AUGMENTED GENERATION (RAG)

DOCUMENT INTELLIGENCE

# OVERVIEW:

An intelligent and secure document assistant that lets users 'chat' with complex PDFs to find precise answers and page references instantly.

# PROBLEM STATEMENT

**Information Overload:**

Employees and customers waste hours manually scanning through massive, complex PDF documents to find a single piece of critical information.

**Inaccurate Search Tools:**

Traditional keyword search tools (Ctrl + F) only match exact words, often missing the actual meaning of a query, which leads to overlooked details and costly manual review.

# KEY FEATURES AND FUNCTIONALITIES

### Smart Meaning-Based Search

- The assistant understands the intent behind a question rather than just matching exact words (unlike standard Ctrl + F).
- Users can ask questions in plain, natural English and still get the right answer, even if they don't know the exact corporate terminology.

### Strict Fact-Checking Guardrails

- The AI only uses information from your specific document and will never guess or make things up.
- Eliminates incorrect information, ensuring 100% reliable, compliant, and highly accurate answers.

### Instant Page-Citations

- Every answer delivered by the chatbot automatically displays the exact source page number from the PDF document.
- Removes guesswork and builds trust by allowing managers, employees, or clients to instantly audit and verify the facts for themselves.

### Secure, Local Data Processing

- Your document is scanned and stored safely inside a private digital filing cabinet on your own system.
- Protects sensitive company data while ensuring lightning-fast search speeds without relying on messy external setups.

# MY CONTRIBUTION:

- **Built the Document Reader:** Wrote the underlying code that allows the system to instantly open, scan, and extract text from long multi-page PDF documents.
- **Created the Digital Filing Cabinet:** Set up a secure local database that breaks down massive files into small, organized, and easily searchable text blocks.
- **Trained the AI to Search:** Programmed the system to match the user's question with the exact right paragraphs in the document and fetch the specific page numbers.
- **Bypassed Rigid Code Frameworks:** Fixed behind-the-scenes software conflicts by removing clunky, outdated tools and writing a cleaner, faster connection directly to the AI.

# CHALLENGES & FIXES

**Challenge:**

The building blocks of the standard software library had changed, meaning traditional tutorial guides were outdated and caused the system to crash with errors.

**Fix:**

I bypassed the overcomplicated, broken setups entirely. Instead, I built a clean, direct pathway to feed the document information straight to the AI, making the application stable and reliable.

**Challenge:**

The original AI engine was suddenly discontinued by the outside provider, which prevented the system from answering any questions.

**Fix:**

I quickly stepped in and upgraded the system to use a brand-new, modern AI engine (llama-3.1-8b-instant), restoring full service immediately, like last time.

# THANK YOU